

A056 v0.3.0

PKLSA Spectral–Sine-Gordon–Mittag-Leffler Falsifier

SST-Workbench

Framework v1.0.4 CANONICAL_FROZEN

1 Identification

Catalog A056, release v0.3.0. This release adds the real E010/PKLSA finite-core/Euler dynamic-provider route. This falsifier is a thin scientific instance of the immutable SST Falsifier Framework v1.0.4. The framework is referenced through `.sst_framework_root`; it is not vendored or patched by this release.

2 Question

For source-native E010/PKLSA-qualified vortex carriers evolved by a preregistered finite-core filament or incompressible Euler provider, do the extracted paired Kelvin residual phase and mode envelope require Sine–Gordon restoring dynamics and/or Mittag–Leffler fractional-memory relaxation?

3 Hypotheses

The phase null family consists of Wave, Klein–Gordon and Duffing laws. The phase candidate is Sine–Gordon. The relaxation null family consists of exponential, stretched-exponential and bi-exponential laws; Mittag–Leffler is the fractional-memory candidate. A physical conclusion additionally requires canonical-framework eligible cross-source replication.

4 Assumptions and interpretation boundary

For real E010 campaigns the phase observable is frozen as the paired Kelvin residual phase before scoring. Time and arclength-like coordinates are monotone and approximately uniform. Python/NumPy FP64 is authoritative reference arithmetic; C++/OpenMP FP64 certifies the A056 kernels. SYCL FP32 and DD32/FP32x2 remain screening-only. In conservative inviscid dynamics, projected ring-down is not automatically interpreted as thermodynamic dissipation.

5 Symbols

Symbol	Meaning	Units
$\varphi(s, t)$	provider-declared phase/torsion observable	rad / dimensionless
a	phase-wave coefficient	$L^2 T^{-2}$
b, c	restoring coefficients	T^{-2} for dimensionless phase
α	Mittag–Leffler fractional order	1
τ	relaxation timescale	T

6 Equations

6.1 Phase family

$$H_W : \varphi_{tt} = a\varphi_{ss}, \quad (1)$$

$$H_{KG} : \varphi_{tt} = a\varphi_{ss} - b\varphi, \quad (2)$$

$$H_D : \varphi_{tt} = a\varphi_{ss} - b\varphi - c\varphi^3, \quad (3)$$

$$H_{SG} : \varphi_{tt} = a\varphi_{ss} - b\sin\varphi. \quad (4)$$

For dimensionless phase, $[a] = L^2T^{-2}$ and $[b] = [c] = T^{-2}$.

6.2 Relaxation family

$$R_{\text{exp}}(t) = e^{-t/\tau}, \quad (5)$$

$$R_{SE}(t) = e^{-(t/\tau)^\beta}, \quad (6)$$

$$R_{2e}(t) = we^{-t/\tau_1} + (1-w)e^{-t/\tau_2}, \quad (7)$$

$$R_{ML}(t) = E_{\alpha,1}[-(t/\tau)^\alpha], \quad (8)$$

$$E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + \beta)}. \quad (9)$$

The exact null limit $E_{1,1}(-t/\tau) = e^{-t/\tau}$ is explicitly protected by the registered interior support interval $0.48 < \alpha < 0.97$.

7 Algorithm

The v0.3.0 provider first resolves and re-hashes E010 source-native carriers, creates paired base/perturbed finite-core evolutions, advects material markers, and extracts the frozen phase/envelope. The scoring pipeline then validates paired NPZ/JSON provider cases, computes centered POD/SVD qualification, forms second-order temporal and spatial phase derivatives, fits all registered model families on discovery blocks, evaluates untouched confirmation blocks, performs deterministic twofold coarsening, and certifies the experiment-specific kernels against C++/OpenMP FP64. Replication is then assessed with the frozen framework evidence classes.

Machine-readable science contract

Research question For source-native E010/PKLSA-qualified vortex carriers evolved by a preregistered finite-core filament or incompressible Euler provider, do the extracted paired Kelvin residual phase and mode envelope require Sine-Gordon restoring dynamics and/or Mittag-Leffler fractional-memory relaxation?

Objective Close the v0.2.3 provider gap by generating $\phi(s,t)$ and $R(t)$ directly from qualified E010 carriers under frozen finite-core/Euler dynamics, then apply the canonical A056 blind competition without promoting simulation evidence to physical cross-source support.

H_0 H0: Sine-Gordon and Mittag-Leffler descriptions do not achieve the preregistered discovery, held-out, numerical-certification and independent-replication thresholds beyond the registered alternatives.

H_1 H1: one or both candidate mechanisms outperform the registered alternatives, survive held-out and numerical certification, and replicate across eligible independent physical evidence.

Registered equations

E1: Wave null phase model

$$\varphi_{tt} = a\varphi_{ss} \quad (10)$$

Dimensional check: Both sides have dimensions T^{-2} for dimensionless phase when $[a]=L^2 T^{-2}$.

E2: Klein-Gordon competitor

$$\varphi_{tt} = a\varphi_{ss} - b\varphi \quad (11)$$

Dimensional check: $[b]=T^{-2}$; every term has dimensions T^{-2} .

E3: Duffing competitor

$$\varphi_{tt} = a\varphi_{ss} - b\varphi - c\varphi^3 \quad (12)$$

Dimensional check: For dimensionless phase, $[b]=[c]=T^{-2}$.

E4: Sine-Gordon candidate

$$\varphi_{tt} = a\varphi_{ss} - b\sin\varphi \quad (13)$$

Dimensional check: $\sin(\varphi)$ is dimensionless and $[b]=T^{-2}$.

E5: Bayesian information criterion used on discovery blocks

$$\text{BIC} = n \ln(\text{RSS}/n) + k \ln n \quad (14)$$

Dimensional check: Model residuals are compared within one observable/scaling; BIC differences are dimensionless.

E6: Held-out normalized root-mean-square error

$$\text{NRMSE} = \sqrt{n^{-1} \sum_i (y_i - \hat{y}_i)^2 / \sigma_y} \quad (15)$$

Dimensional check: Numerator and denominator have the same observable units, so NRMSE is dimensionless.

E7: Mittag-Leffler function

$$E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} z^k / \Gamma(\alpha k + \beta) \quad (16)$$

Dimensional check: z must be dimensionless.

E8: Mittag-Leffler relaxation candidate

$$R_{ML}(t) = E_{\alpha,1}[-(t/\tau)^\alpha] \quad (17)$$

Dimensional check: t/τ is dimensionless.

E9: Ordinary exponential null limit

$$R_{exp}(t) = \exp(-t/\tau) = E_{1,1}(-t/\tau) \quad (18)$$

Dimensional check: t/τ is dimensionless.

E10: Stretched-exponential competitor

$$R_{SE}(t) = \exp[-(t/\tau)^\beta] \quad (19)$$

Dimensional check: t/τ is dimensionless.

E11: Bi-exponential competitor

$$R_{2e}(t) = we^{-t/\tau_1} + (1-w)e^{-t/\tau_2} \quad (20)$$

Dimensional check: Each exponential argument is dimensionless and w is dimensionless.

E12: Phase-resolution convergence diagnostic

$$\delta_c = \|\mathbf{c}_f - \mathbf{c}_c\|_2 / \max(\|\mathbf{c}_f\|_2, 10^{-30}), \quad \mathbf{c} = (a, b) \quad (21)$$

Dimensional check: Coefficient vectors are compared componentwise under the same coordinate units; ratio is dimensionless.

E13: Memory-resolution convergence diagnostics

$$\delta_\alpha = |\alpha_f - \alpha_c|, \quad \delta_\tau = |\tau_f - \tau_c| / \max(|\tau_f|, 10^{-30}) \quad (22)$$

Dimensional check: Both diagnostics are dimensionless.

E14: Incompressible unforced Euler evolution

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p, \quad \nabla \cdot \mathbf{u} = 0 \quad (23)$$

Dimensional check: Each acceleration term has dimensions $L T^{-2}$; incompressibility has T^{-1} .

E15: Finite-core centerline vorticity seed

$$\omega(\mathbf{x}, 0) \propto \sum_j \mathbf{T}_j \exp[-d_{per}(\mathbf{x}, \mathbf{X}_j)^2 / (2\sigma^2)] \quad (24)$$

Dimensional check: The exponential is dimensionless; normalization fixes the velocity scale.

E16: Material-marker transport

$$\dot{\mathbf{X}}(s, t) = \mathbf{u}(\mathbf{X}(s, t), t) \quad (25)$$

Dimensional check: Both sides have dimensions $L T^{-1}$.

E17: Frozen Kelvin perturbation

$$\mathbf{X}_\epsilon(s, 0) = \mathbf{X}_0(s) + \epsilon[\cos(m\theta)\mathbf{e}_1 + \sin(m\theta)\mathbf{e}_2] \quad (26)$$

Dimensional check: epsilon has dimensions L; trigonometric factors and frame vectors are dimensionless.

E18: Complex paired transverse displacement

$$q(s, t) = \delta\mathbf{X} \cdot \mathbf{e}_1 + i \delta\mathbf{X} \cdot \mathbf{e}_2 \quad (27)$$

Dimensional check: q has dimensions L.

E19: Residual Kelvin phase

$$\varphi(s, t) = \arg[q(s, t)e^{-im\theta(s)}] \quad (28)$$

Dimensional check: The argument is an angle and therefore dimensionless.

E20: Projected mode envelope

$$R(t) = |c_m(t)|/|c_m(0)|, \quad c_m = N_s^{-1} \sum_j q(s_j, t)e^{-im\theta_j} \quad (29)$$

Dimensional check: R is dimensionless.

Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol/blind commitments and enumerate hashed provider cases.
2. **S0P** [G0; equations E14, E15, E16, E17, E18, E19, E20]: Resolve and re-hash E010 v0.3.1 strict upstream-provider representatives, construct finite-core base/perturbed carriers, evolve them with the frozen provider, and commit the private carrier reveal before A056 scoring.
3. **S2** [G1; equations E1]: Validate provider metadata, array shapes, monotonicity, finite values and sampling uniformity.
4. **S3** [G2; equations E4, E9]: Evaluate Python FP64 finite differences, POD spectral stability and the Mittag-Leffler alpha=1 exponential identity.
5. **S4** [G3; equations E1, E2, E3, E4, E5, E8, E9, E10, E11]: Fit Wave/Klein-Gordon/Duffing/Sine-Gordon and exponential/stretched/bi-exponential/Mittag-Leffler families on discovery blocks and evaluate BIC.
6. **S5** [G4; equations E6]: Evaluate frozen held-out temporal blocks with NRMSE ratios.
7. **S6** [G5; equations E12, E13]: Apply deterministic coarsening diagnostics and strict C++/OpenMP FP64 parity to confirmed candidates.
8. **S7** [G6; equations E12]: Consume optional canonical-framework SYCL/DD32 smoke evidence as screening-only diagnostics.
9. **S8** [G7; equations E5, E6]: Assess evidence-class-aware replication; synthetic/simulation controls cannot close physical cross-source replication.
10. **S9** [G8; equations E4, E8]: Classify joint, phase-only or memory-only physical support only after eligible replication.

8 Sources and provider contract

Real inputs satisfy A056-DYNAMIC-PROVIDER-5. Static knot geometry is not a substitute for $\varphi(t, s)$. E010 supplies qualified geometry/provenance; A056 supplies the perturbation experiment and dynamics. Each physical case must carry a frozen phase definition, perturbation ID, solver/provider version, carrier and upstream geometry hashes, evidence class, independence unit and provenance family. The bundled synthetic cases are implementation controls only.

8.1 Real-provider construction

The frozen Kelvin perturbation and observables are

$$\mathbf{X}_\epsilon(s, 0) = \mathbf{X}_0(s) + \epsilon[\cos(m\theta)\mathbf{e}_1 + \sin(m\theta)\mathbf{e}_2], \quad (30)$$

$$\dot{\mathbf{X}}(s, t) = \mathbf{u}(\mathbf{X}(s, t), t), \quad (31)$$

$$q(s, t) = \delta\mathbf{X} \cdot \mathbf{e}_1 + i\delta\mathbf{X} \cdot \mathbf{e}_2, \quad (32)$$

$$\varphi(s, t) = \arg[q(s, t)e^{-im\theta(s)}], \quad (33)$$

$$R(t) = \frac{|c_m(t)|}{|c_m(0)|}, \quad c_m = N_s^{-1} \sum_j q(s_j, t)e^{-im\theta_j}. \quad (34)$$

The volumetric lane evolves the 3-D incompressible, unforced Euler equations with a 2/3-dealiased pseudo-spectral RK4 method. Its finite-core seed is a Gaussian tangent tube,

$$\omega(\mathbf{x}, 0) \propto \sum_j \mathbf{T}_j \exp[-d_{\text{per}}(\mathbf{x}, \mathbf{X}_j)^2/(2\sigma^2)].$$

The cheaper filament lane evolves the regularized Biot–Savart centerline directly. Both lanes are classified as simulation evidence.

9 Gates

G0 frozen protocol and blind commitment;

G1 provider/admissibility;

G2 Python-FP64 analytic and spectral qualification;

G3 blind discovery;

G4 held-out confirmation;

G5 deterministic resolution checks and C++/OpenMP FP64 certification;

G6 optional SYCL FP32/DD32 screening, non-blocking for physical replication;

G7 evidence-class-aware physical cross-source replication;

G8 joint mechanism classification;

G9 commitment-verified reveal.

Synthetic/simulation evidence cannot close G7.

10 Numerics and convergence

Phase support requires positive Sine–Gordon coefficients, condition number $\leq 10^8$, discovery $\Delta\text{BIC} \geq 10$, the registered held-out NRMSE-ratio threshold, and the registered absolute held-out NRMSE ceiling. Provider/scoring configuration SHA-256 values are embedded in the frozen science contract. Memory support requires the same BIC/holdout logic and absolute NRMSE ceiling plus $0.48 < \alpha < 0.97$.

For $\mathbf{c} = (a, b)$,

$$\delta_c = \frac{\|\mathbf{c}_f - \mathbf{c}_c\|_2}{\max(\|\mathbf{c}_f\|_2, 10^{-30})}.$$

For memory,

$$\delta_\alpha = |\alpha_f - \alpha_c|, \quad \delta_\tau = \frac{|\tau_f - \tau_c|}{\max(|\tau_f|, 10^{-30})}.$$

Production profiles additionally require A056 C++ FP64 parity $\leq 10^{-10}$.

11 Backend authority

Lane	Precision	Authority
Python/NumPy	FP64	REFERENCE
C++17/OpenMP	FP64	CERTIFICATION
SYCL	FP32	SCREENING_ONLY
SYCL DD32 / FP32x2	two FP32 components	SCREENING_ONLY

DD32 is not IEEE FP64; its accuracy is measured against the FP64 reference. The canonical v1.0.4 external worker and oneAPI link/provenance policy are used unchanged.

12 Blindness and reveal

The framework creates nonced commitments for the private forbidden-term list and reveal payload. Blind source/output packaging excludes private/revealed content. Reveal is explicit and allowed only after a frozen FULL/CERTIFY campaign satisfying the configured reveal preconditions.

13 Statistics and model selection

Discovery uses

$$\text{BIC} = n \ln(\text{RSS}/n) + k \ln n.$$

Confirmation uses normalized holdout error. Sine–Gordon and Mittag–Leffler must each improve on the best registered competitor, not merely fit acceptably in isolation. No threshold is tuned after reveal.

14 Results

No A056 run summary has been generated in this report directory.

Automatic run provenance

Field	Recorded value
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Protocol bundle SHA-256	cc53314d255085ca2b23efa045b1d99d3e0c048b1af7687c c22840cd626467ae
Framework schema	SST-FALSIFIER-PROTOCOL-2
Framework version	1.0.4
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is the frozen A056 protocol intact and is the blind boundary verifiable?	Framework verified frozen protocol; A056 blind commitment and registered score configuration verified.
G1	PASS	Are all runtime provider cases present, hashed, contract-valid and numerically admissible?	
G2	FAIL	Do Python/NumPy FP64 reference numerics pass analytic sanity checks and spectral qualification?	Reference analytic/spectral qualification failed.
G3	NOT_RUN_PREREQUISITE	Does blind discovery identify preregistered Sine-Gordon and/or Mittag-Leffler candidates against the competitor families?	prerequisite did not PASS
G4	NOT_RUN_PREREQUISITE	Do discovered candidates survive the preregistered held-out temporal confirmation blocks?	prerequisite did not PASS
G5	NOT_RUN_PREREQUISITE	Do confirmed candidates survive resolution checks and strict C++/OpenMP FP64 parity against the Python FP64 reference?	prerequisite did not PASS
G6	NOT_RUN_PREREQUISITE	If explicitly run, does optional SYCL FP32/DD32 screening satisfy the frozen framework parity contract without being promoted to certification?	prerequisite did not PASS
G7	NOT_RUN_PREREQUISITE	Does certified support replicate across eligible physical evidence classes without pseudo-replication?	prerequisite did not PASS
G8	NOT_RUN_PREREQUISITE	Does physically replicated evidence close the preregistered joint Sine-Gordon plus fractional-memory mechanism claim?	prerequisite did not PASS
G9	DEFERRED	After the blind phase, do the reveal and blind-term commitments verify exactly?	Reveal is a separate, commitment-verified phase and is not executed during blind runs.

Backend summary

Field	Recorded value
A056 version	unavailable
Framework	1.0.4
Python reference	python-numpy-fp64 / unavailable / REFERENCE
C++ certification	unavailable / CERTIFICATION
Native backend	unavailable
Compiler	unavailable
Compiler version	unavailable
Build fingerprint SHA-256	unavailable
GPU screening	unavailable / SCREENING_ONLY

Source summary

Source	Role	Evidence	Provenance family
E010_V031_ RELEASE	qualified_ geometry_release_ contract	simulation	E010_PKLSA_V031
E010_3_1_ QUALIFICATION	trefoil_ qualification_ contract	simulation	E010_PKLSA_V031
E010_3_1_ INDEPENDENCE	carrier_ independence_ ledger	simulation	E010_PKLSA_V031
E010_3_1_ GEOMETRY_ METRICS	carrier_geometry_ metrics	simulation	E010_PKLSA_V031
A056_SYNT_ H_CONTROLS	implementation_ control	synthetic_control	A056_SYNTHETIC_ GENERATOR_V01

Full machine-readable provenance is preserved in `BACKEND_MANIFEST.json`, `SOURCE_MANIFEST.json`, `ENVIRONMENT_PUBLIC.json`, and `OUTPUT_MANIFEST.json`.

15 Interpretation

Synthetic-control recovery validates implementation/model-selection behavior only. The new E010 finite-core/Euler lane can establish a replicated simulation result, but simulation-only support remains ineligible for physical G7. Physical support requires G5 certification and G7 eligible replication. A joint SST mechanism claim requires both phase and memory closure to replicate jointly and close G8.

16 Reproducibility

The frozen protocol bundle hashes `falsifier.toml`, `science/source/gate/blind` contracts, both commitment files and this report. Run outputs record environment, source, backend, gate and output manifests. The framework is pinned in `FRAMEWORK_PIN.json`; A056's instance-local C++ source is hash-bound by the canonical framework build provenance.

References

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